

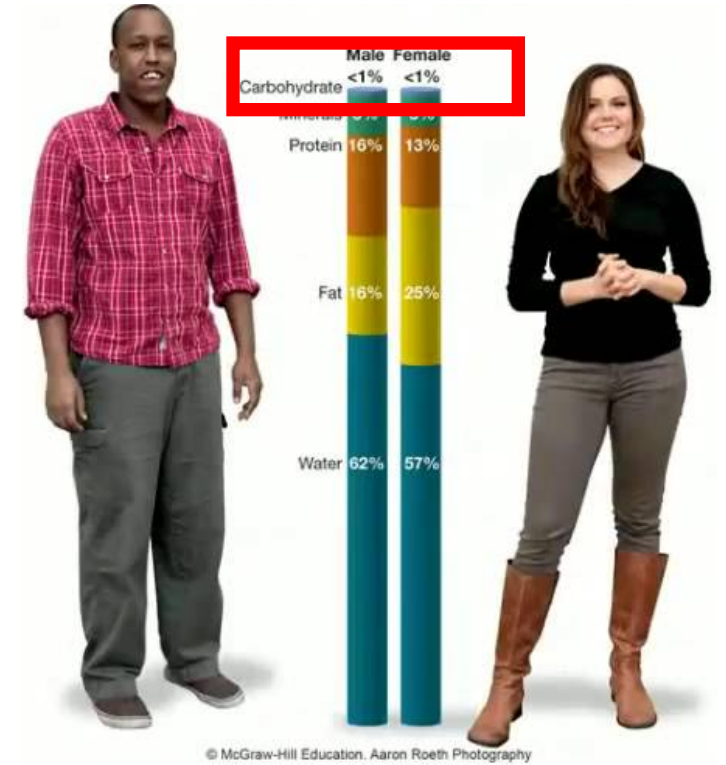
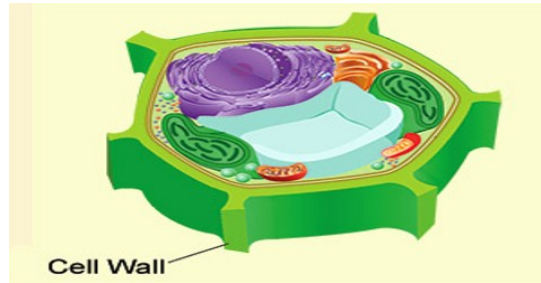
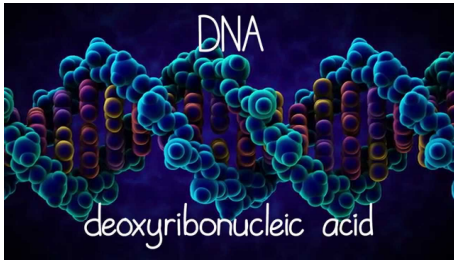
Carbohydrates

Part 1

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BSBE, IIT Guwahati₁

Carbohydrates

- Main source of **ENERGY** (₹ ATP) in all living organism
- Serves as **structural** components
- (e.g, DNA: **RIBOSE**, Plant Cell Wall : **CELLULOSE**)



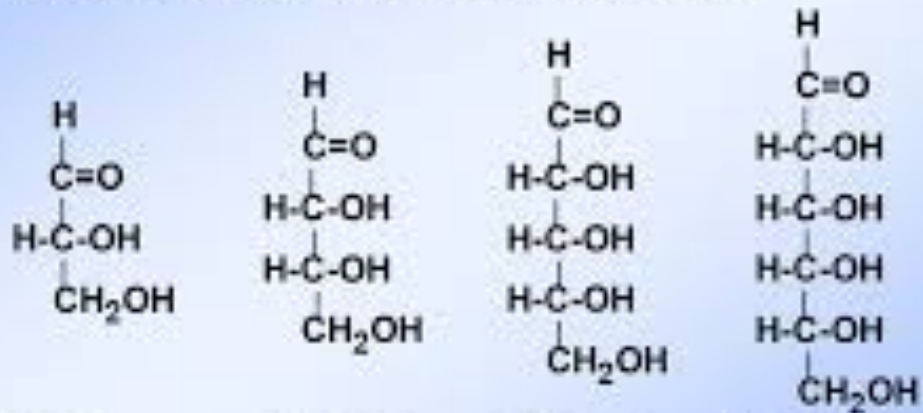
- Carbohydrates contains (CARBON , HYDROGEN, OXYGEN)
- MOLAR RATIO= (1:2:1)
- 4 types of carbohydrates
 - Monosaccharide
 - Disaccharide
 - Oligosaccharide
 - polysaccharide

**Greek work
SACCHARIDE = SUGAR**

Monosaccharides (SIMPLE SUGAR)

Monosaccharide classifications

Number of carbon atoms in the chain



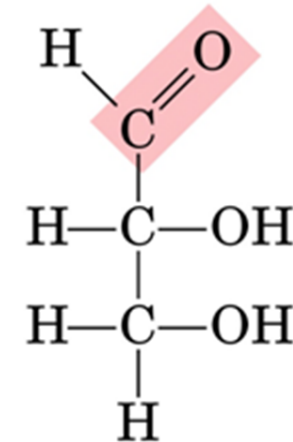
triose

tetrose

pentose

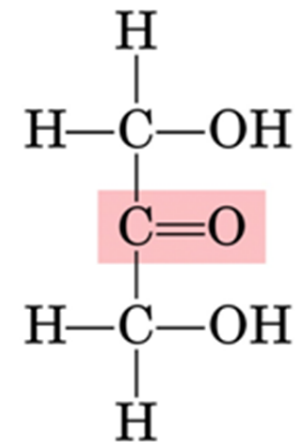
hexose

Can be either aldose or ketose sugar.



Aldose

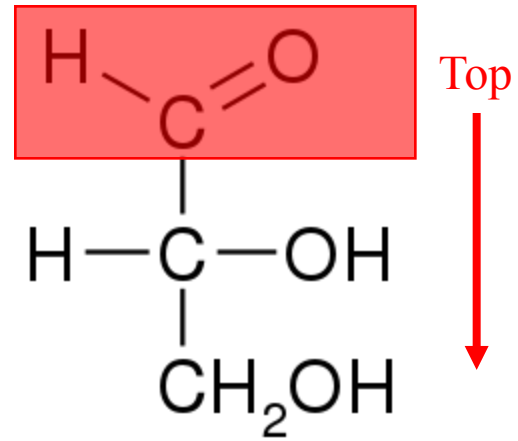
**Aldotriose
/ Glyceraldehyde**



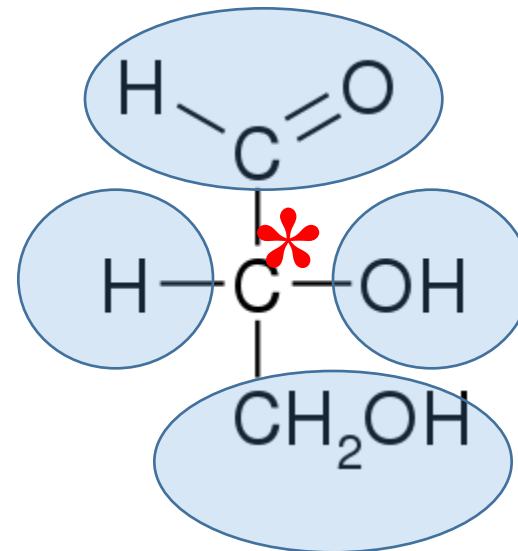
Ketose

**Ketotriose
/ Dihydroxyacetone**

- Aldehyde or ketone at the **top** and the rest of the linear monosaccharides downwards



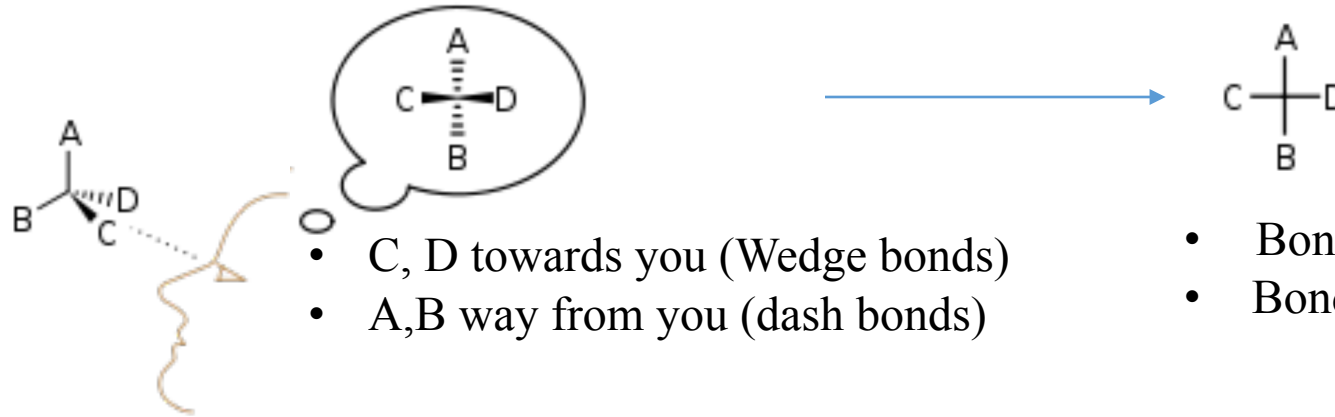
- Chiral center (*) and stereochemistry



Monosaccharides

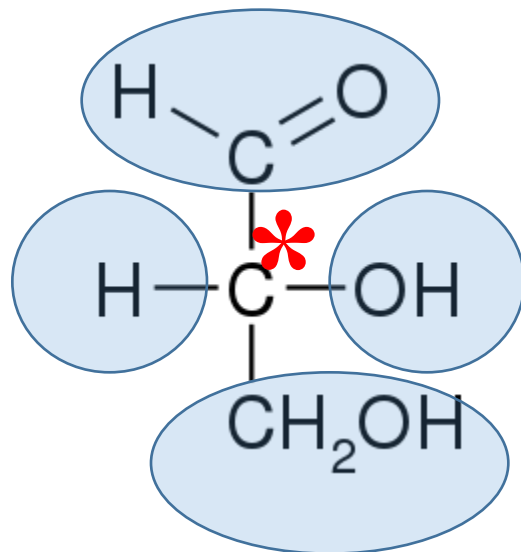
Convention for drawing linear Monosaccharides

- Fischer projection

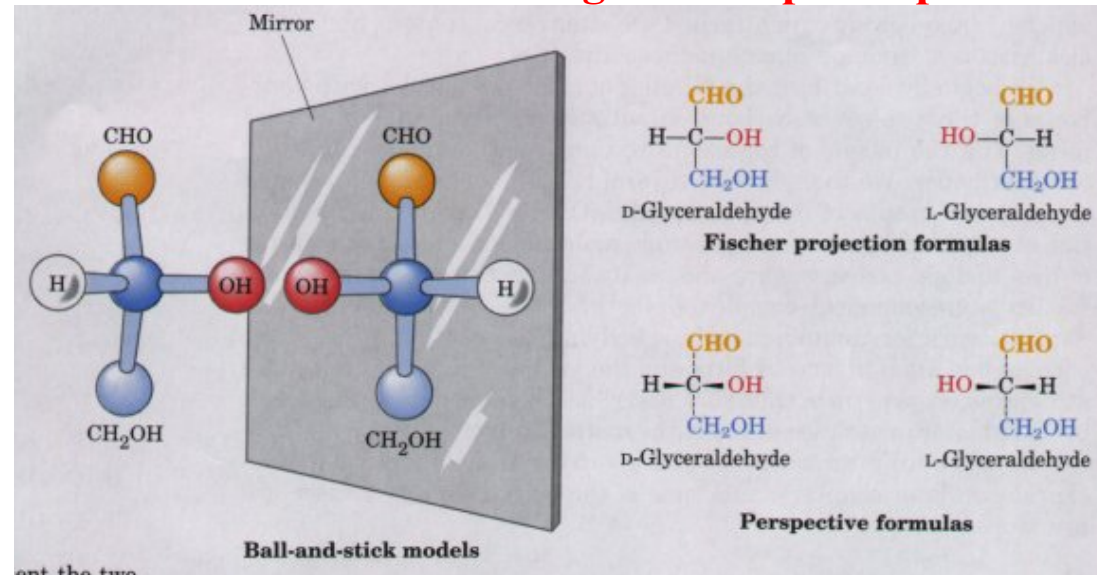


- Bond angles are NOT 90° .
- Bond angles are 109.5°

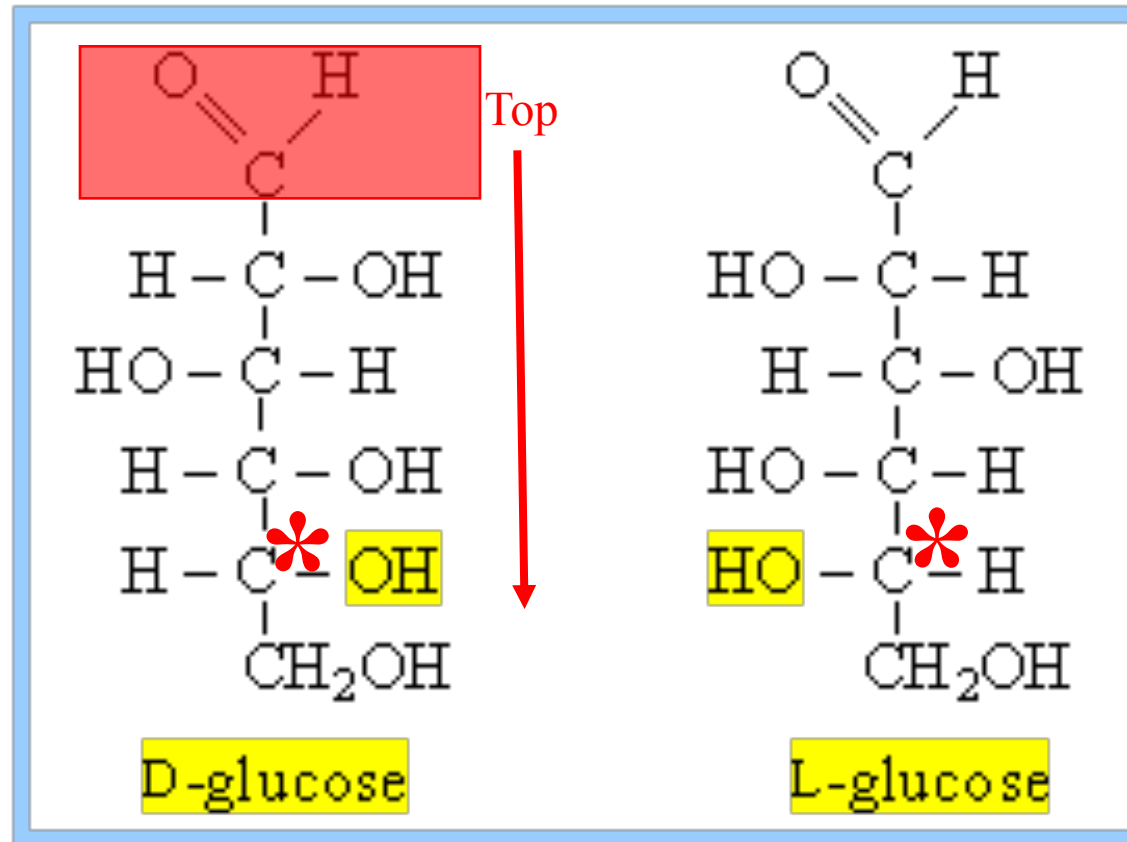
- Chiral center (*) and stereochemistry



Enantiomers : Mirror images not super imposable

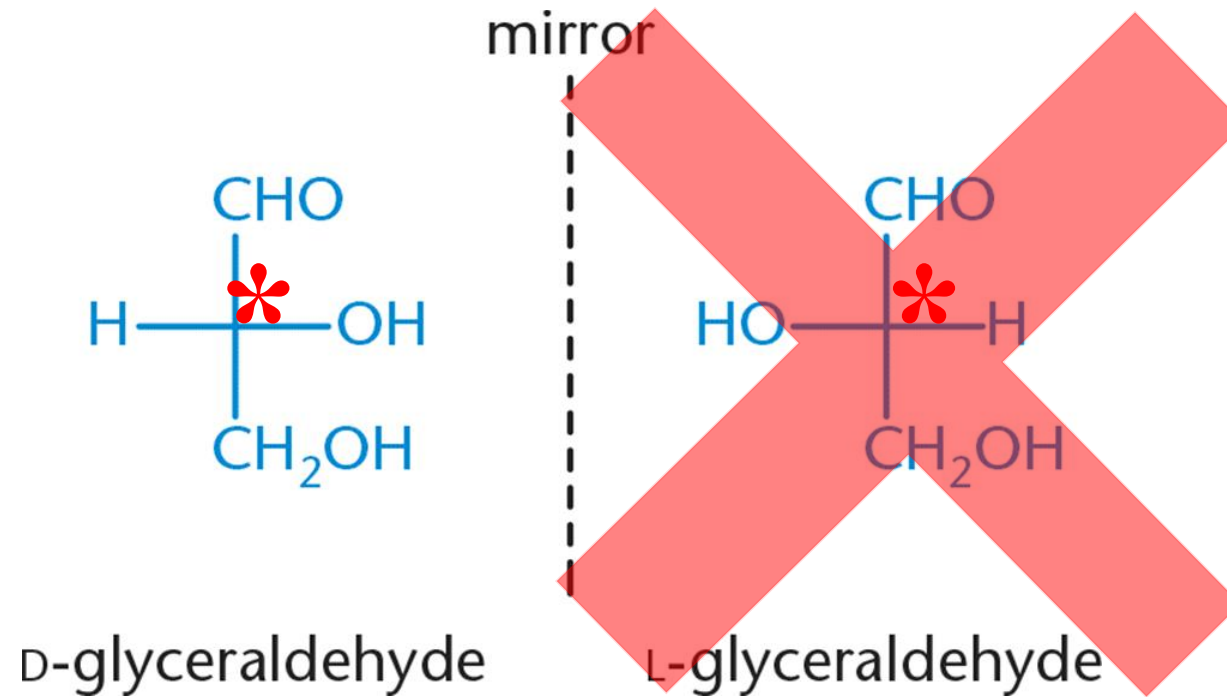


- Stereochemistry of linear monosaccharides with more number of carbon atoms are defined as “**Chiral center at the bottom farthest away from the carbonyl**”



Enantiomers

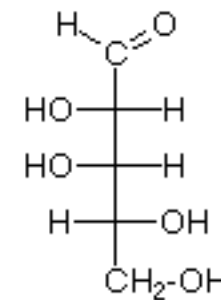
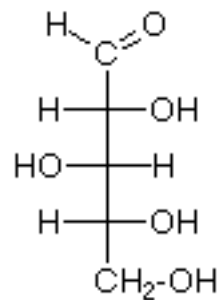
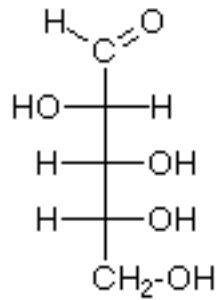
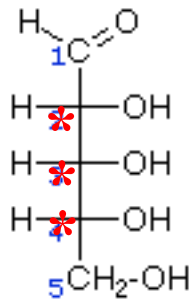
- Aldotriose / Glyceraldehyde is the simplest sugar/monosaccharides
- Only one chiral center (*).
- Terminology D, L isomer terminology based on that



- Nature prefer only “D” isomer (*Reason not known*)
- **Unless specified explicitly, ASSUME “D” sugars always**

n chiral center = 2^n Stereoisomers

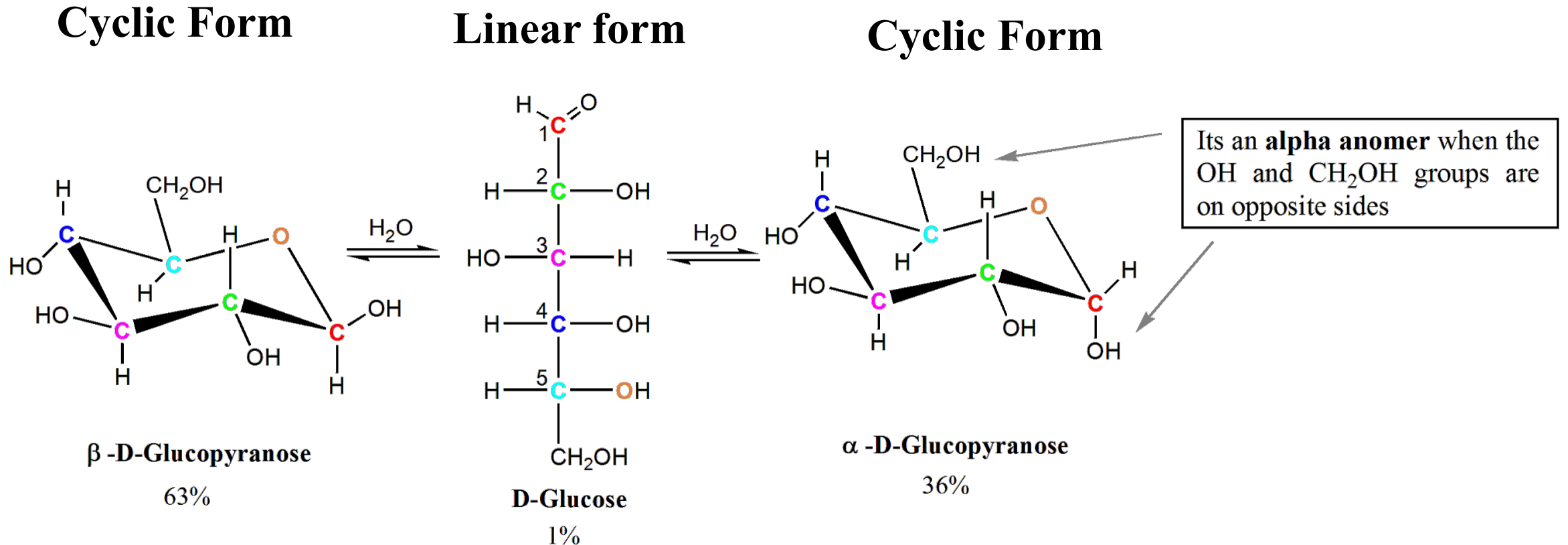
3 chiral center



+ 4 L enantiomers

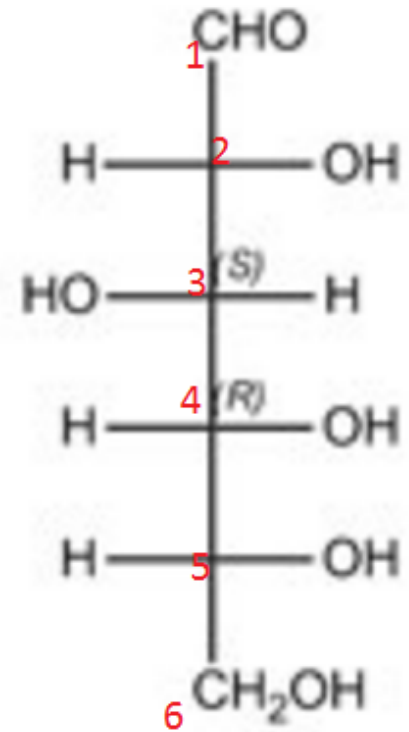
3 chiral center = $2^3 = 8$ Stereoisomers of Aldopentoses

Sugars are in equilibrium (Linear $\rightleftharpoons$ Cyclic) forms

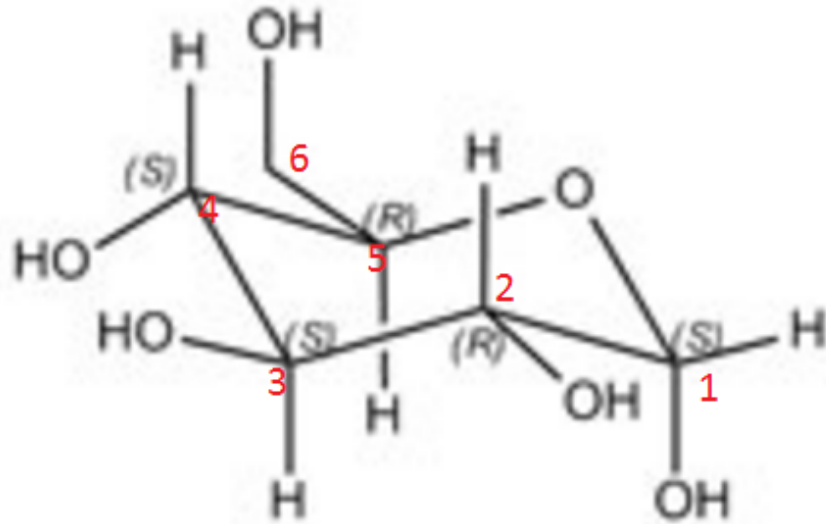


Alcohol attack aldehyde, Intramolecular cyclization...

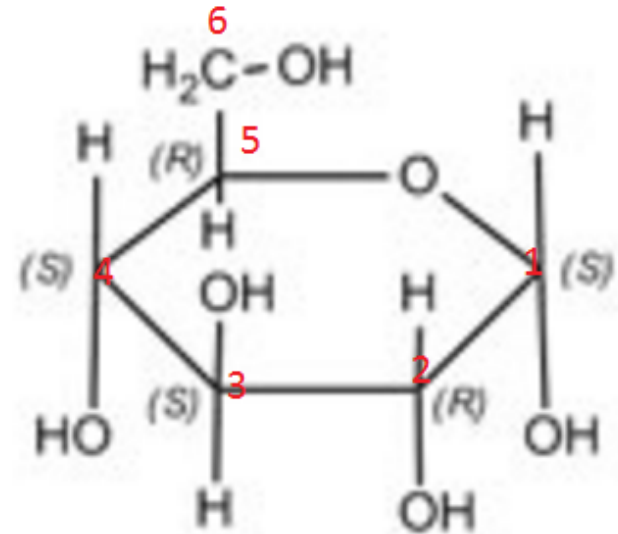
Howarth projection



Fischer projection



Perspective projection



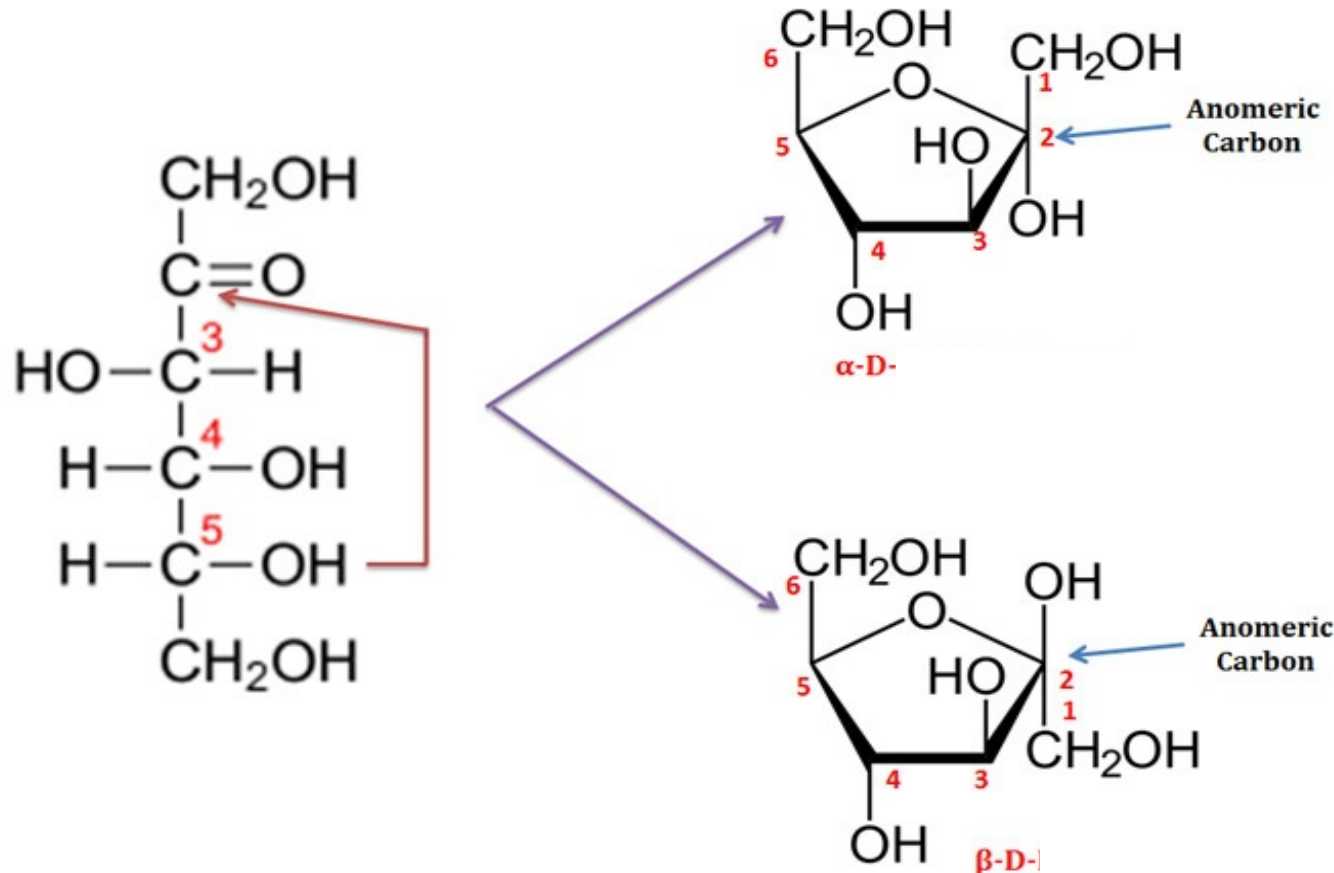
Haworth projection

Sugars are in equilibrium (Linear $\rightleftharpoons$ Cyclic) forms

Cyclic Structure of Fructose

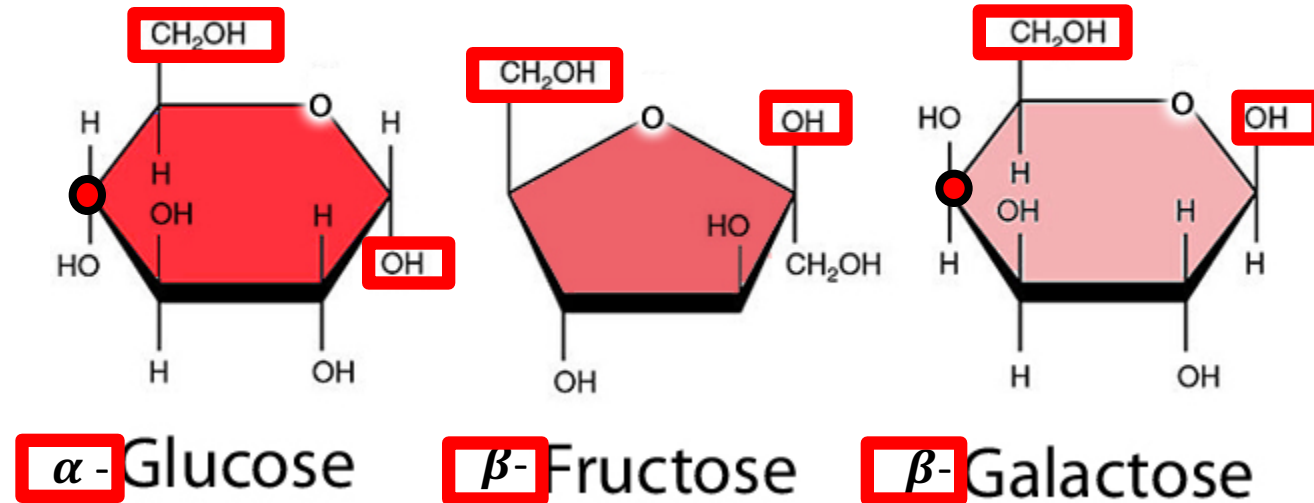
- As a ketohexose, fructose forms a 5-membered ring when the hydroxyl on C-5 reacts with the carbonyl on C-2

Formation of Cyclic Structure of Fructose



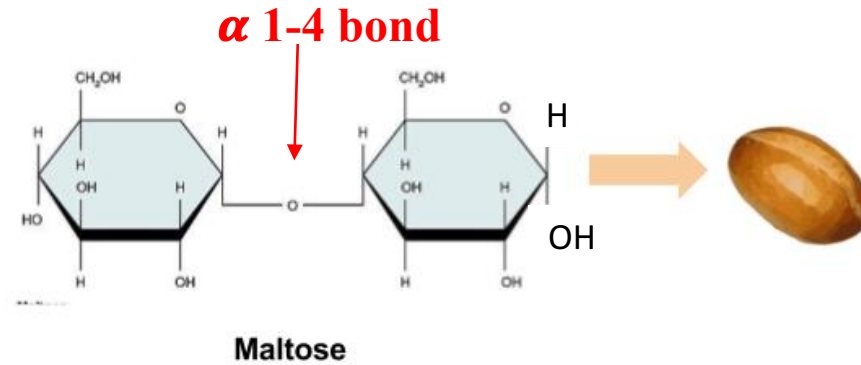
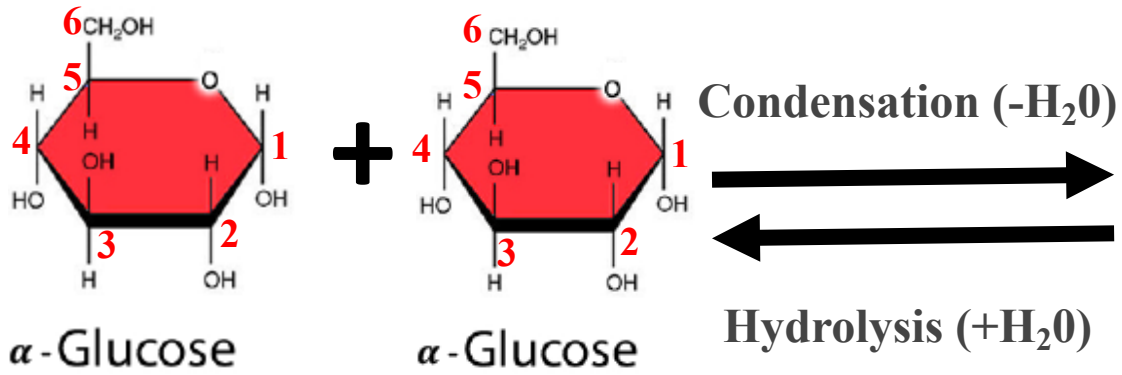
Three main Monosaccharides from HUMAN DIET

Monosaccharides

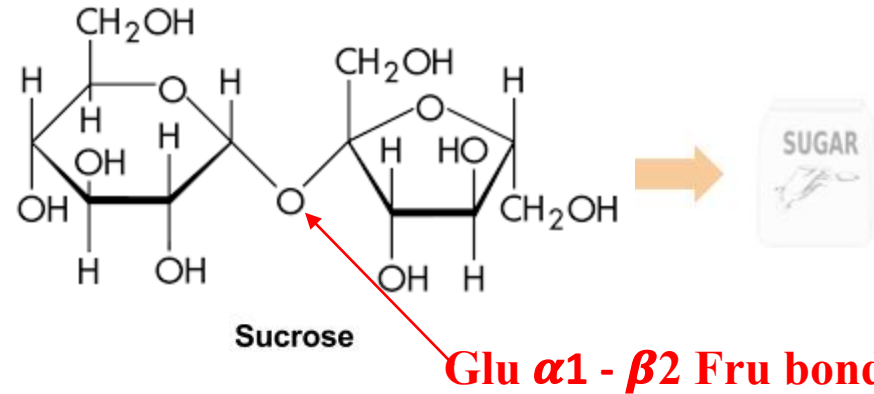
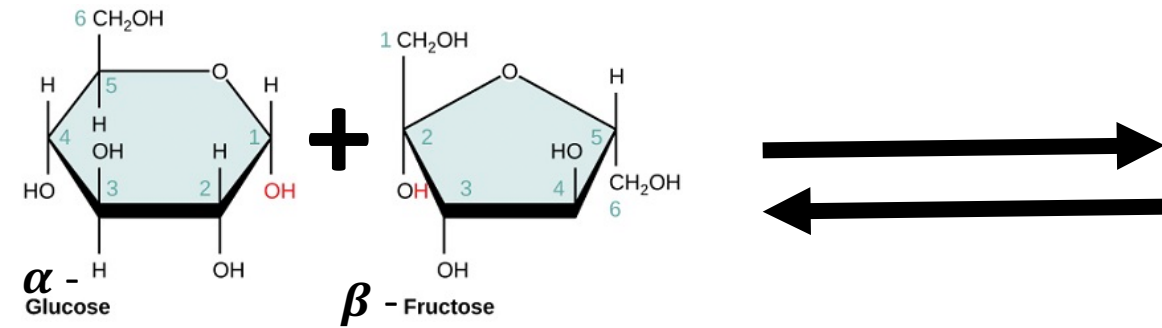


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Disaccharide (Made up of 2 monosaccharides)

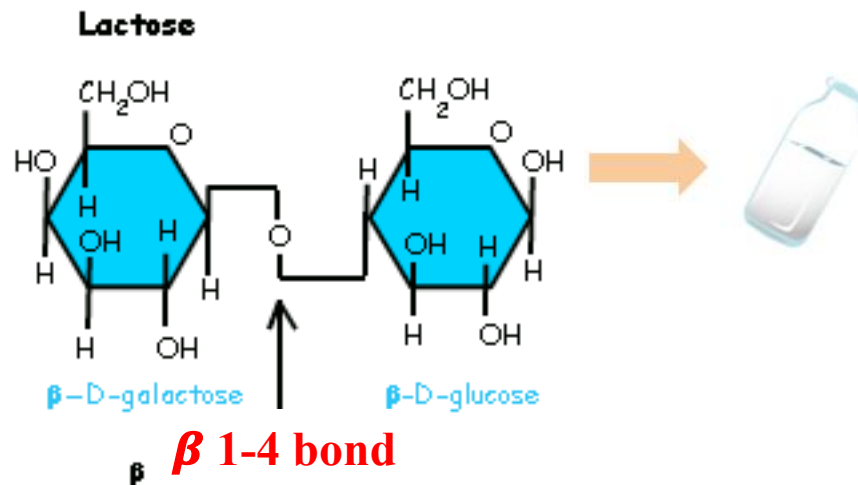


Maltose is formed by two glucose molecules. Maltose is responsible for a slightly sweet taste you experience when chewing bread holding it in your mouth.



Sucrose is formed by linking glucose to fructose. This disaccharide is commonly known as table sugar.

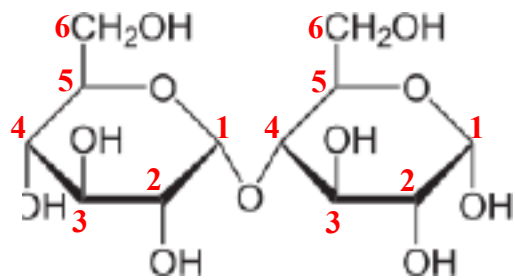
MADE BY PLANTS
NOT ANIMALS



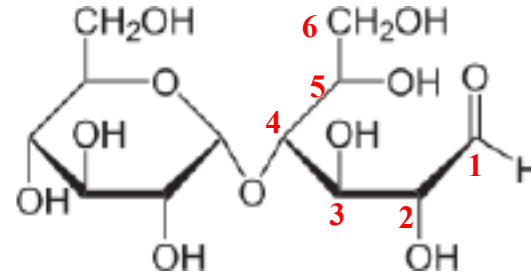
Lactose is formed by the linking of galactose to glucose. This disaccharide is most commonly found in milk products.

Non-reducing sugars

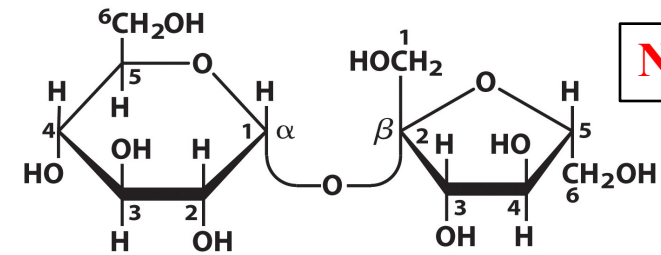
Do not have an OH group attached to the anomeric carbon so they cannot reduce other compounds.



Maltose



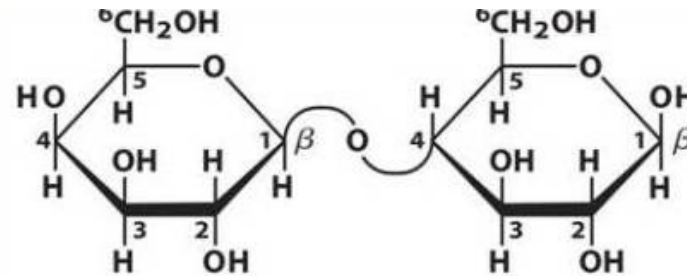
Reducing sugars



Sucrose

α -D-glucopyranosyl β -D-fructofuranoside
Glc(α 1 $\leftrightarrow$ 2 β)Fru

Non-reducing sugars



Lactose (β form)

β -D-galactopyranosyl-(1 $\rightarrow$ 4)- β -D-glucopyranose
Gal(β 1 $\rightarrow$ 4)Glc

Reducing sugars

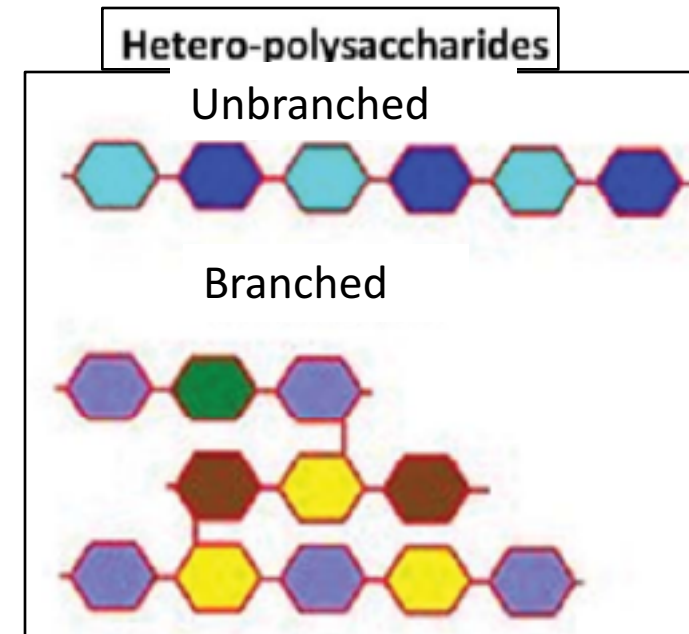
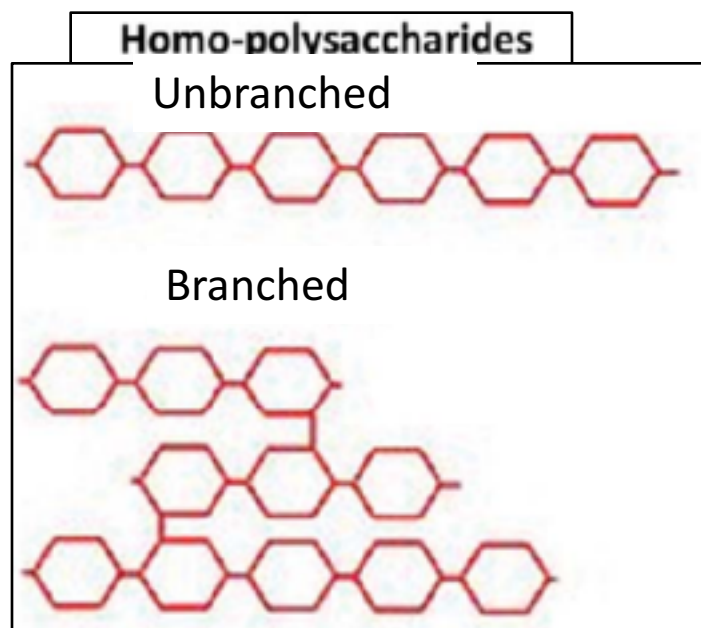
Oligosaccharide (chains of < 20 monosaccharides)



> 20 monosaccharides

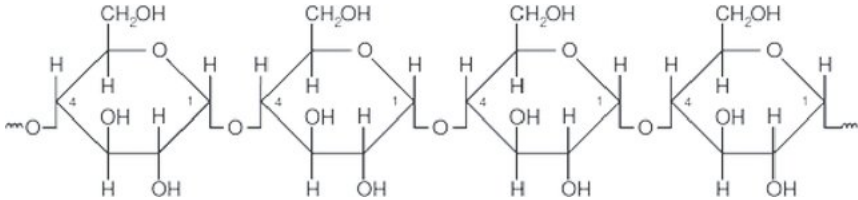
Polysaccharide (COMPLEX SUGER)

**MOST CARBOHYDRATES IN NATURE OCCUR AS POLYSACCHARIDES.
ALSO KNOWN AS GLYCANS**



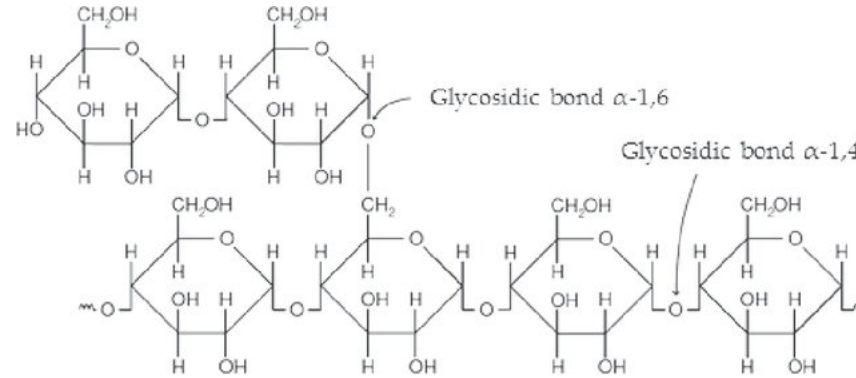
Homo-polysaccharide (Example)

Starch (Storage form in plants) : Chain of Glucose



Unbranched Glucose

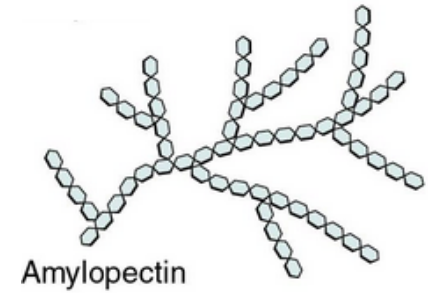
With α 1-4 glycosidic bond = **Amylose**



Branched Glucose with α 1-4 + α 1-6

Glycosidic bond = **Amylopectin**

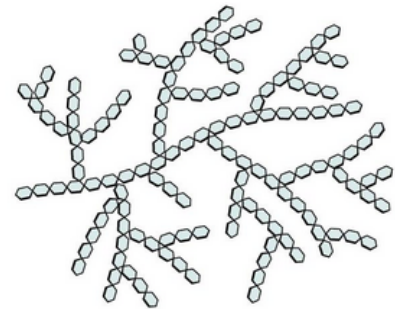
Branched points occur every 24-30 Glucose residues



Amylopectin

Glycogen (Storage form in animals) : Chain of Glucose same as starch

Branched points occur every 8-12 Glucose residues

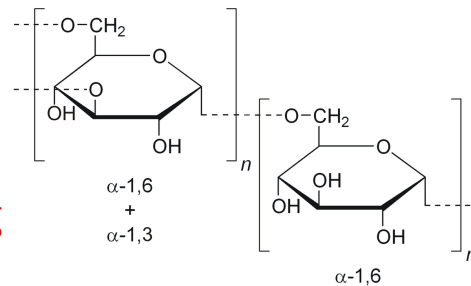


Glycogen

Dextran (produced by bacteria, yeast) :

Made up α 1-6 + α 1-3 (branching).

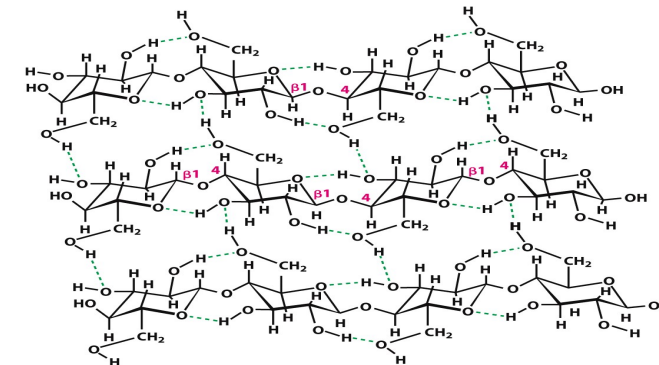
May also contain α 1-2 + α 1-4 branching



Cellulose (Structural component of plants, CELL WALL) :

Unbranched Glucose (> 1000) chains with β 1-4 linkage,

On top of each other held together by H-bonds



Cellulose
(β -1,4 linkages)

Cellulose (Structural component of plants,
CELL WALL) :

**HUMANS DO NOT HAVE ENZYME THAT
CAN HYDROLYZE β 1-4 GLYCOSIDIC BOND:
HENCE CAN NOT BE DIGESTED BY HUMAN**

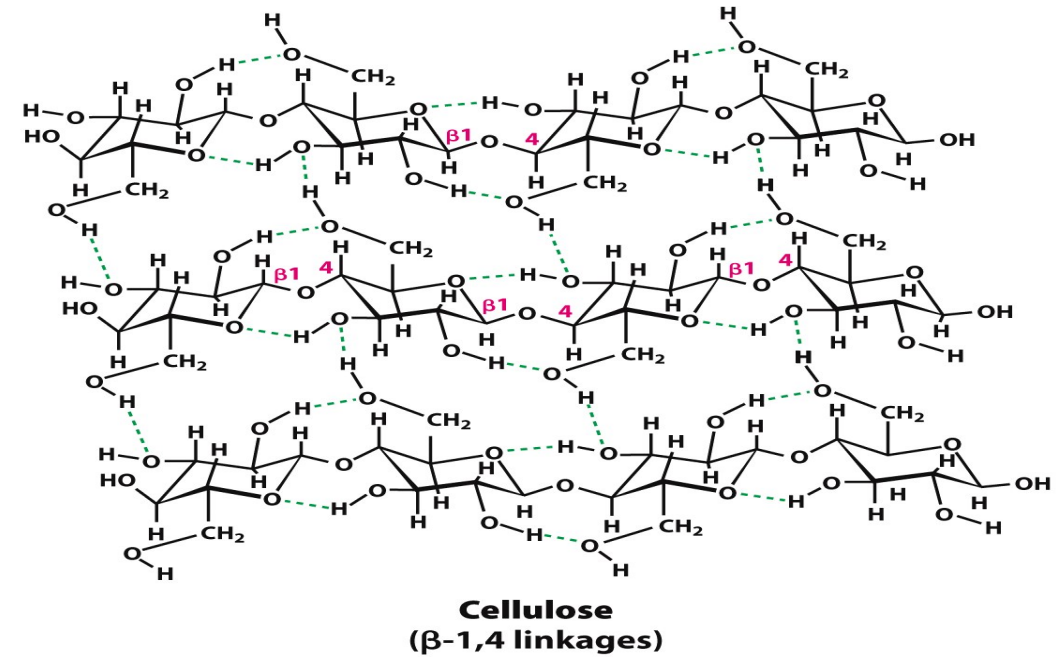
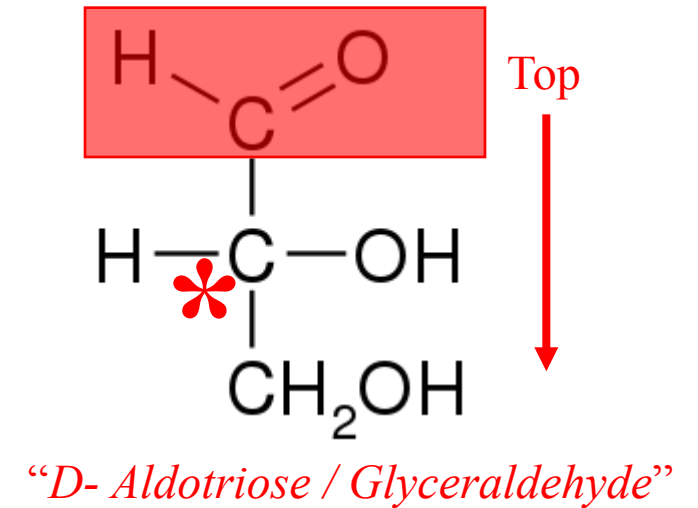
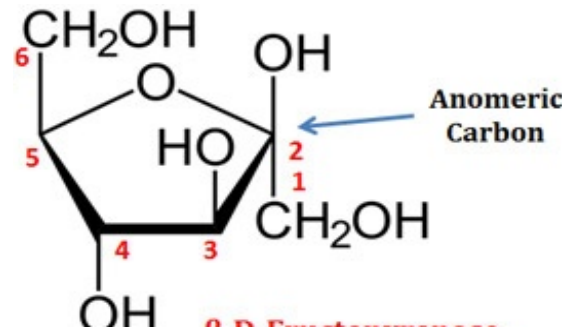
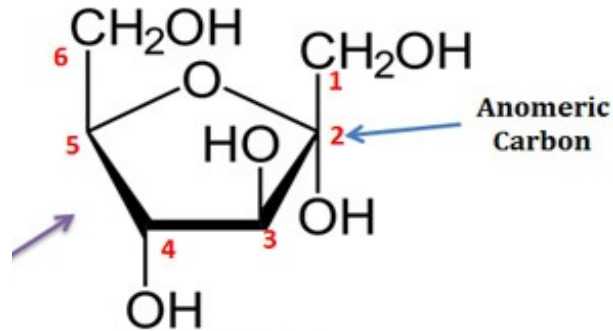


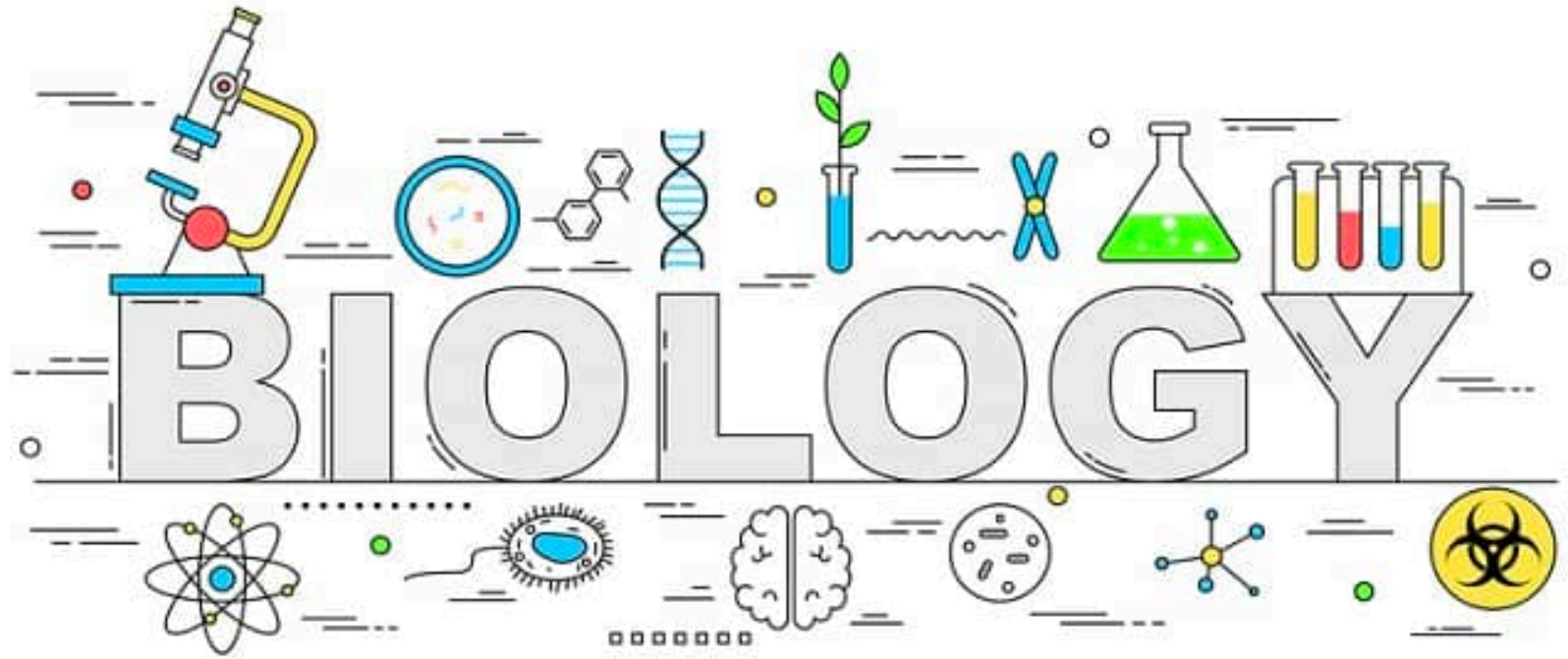
Figure 11.14 part 1
Biochemistry, Seventh Edition
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Take Home

- Carbohydrate = Sugar (Mono/Di/Oligo/Poly –Saccharide)
- Represent saccharide = Aldehyde or ketone at the **Top** and the rest of the linear monosaccharides downwards
- Chiral center (*) farthest away from the carbonyl define “**Stereochemistry (D/L)**”
- Sugars are in equilibrium (**Linear** $\rightleftharpoons$ **Cyclic**) forms
- Cyclic form produces “ α ” and “ β ” anomers.



- Disaccharide : Maltose (Bread), Sucrose (Sugar), Lactose (milk)
- Non reducing sugar : **Do not have an OH group** attached to the **anomeric carbon**
- Starch and Glycogen are energy storage form in plant and animals respectively (Branched glucose: α 1-4+ α 1-6)
- Cellulose: Unbranched Glucose chains with β 1-4 linkage. CAN NOT BE DIGESTED BY HUMAN



Next: Carbohydrates (Part 2)